

Research
Paper

Digital Society
Programme

October 2025

The case for expanding digital public infrastructure

How open, scalable technology can serve citizens, preserve sovereignty and save money

Rowan Wilkinson, Alex Krasodonski and Isabella Wilkinson



Chatham House, the Royal Institute of International Affairs, is a world-leading policy institute based in London. Our mission is to help governments and societies build a sustainably secure, prosperous and just world.

Contents

	Summary	2
01	Introduction	4
02	What is DPI?	7
03	Sovereignty	10
04	Economic opportunity	14
05	Resilience and security	18
06	Public sector modernization	21
07	Global DPI collaboration	24
08	Conclusion and recommendations	27
	Annex 1	32
	About the authors	33
	Acknowledgments	34

Summary

-
- Digital services are critical to the management of government budgets, the modernization of societies and economic growth. This transformation has the potential to boost efficiency, reduce spending, and improve accessibility and inclusion for citizens. However, to date, efforts in this direction have been characterized by high costs, long-term dependence on foreign service providers and siloed projects that result in duplication and waste. Creating a digital society that can unlock new public goods requires an urgent and comprehensive reset of strategic thinking and approach.
 - Digital public infrastructure (DPI) is a way of building society-scale digital systems that can overcome some of the past issues encountered in the digitalization process by adhering to a few key principles. These include only initiating open, public-interest digital solutions that are interoperable with other systems, that can be built upon and that enable the secure sharing of data. Common examples of DPI include digital identity systems, digital payment systems and data exchange systems. The term DPI refers both to the approach and principles behind open, interoperable digitalization, and to the resulting digital systems and platforms that add infrastructural capability.
 - The principles of DPI offer five key benefits for governments looking to build robust digital foundations:
 - **Sovereignty:** DPI approaches aim to counter foreign dependencies and vendor lock-in, while allowing more scope for national principles and values to shape and inform society-scale technology.
 - **Economic opportunity:** By replacing expensive, proprietary and siloed digital services with open, interoperable components that can share data, DPI can increase bureaucratic efficiency, boost access to credit, improve tax collection, lower costs for businesses, and provide a foundation for local innovation.
 - **Resilience and security:** DPI can safeguard national security by reducing reliance on foreign-controlled proprietary systems, as well as by ensuring that critical digital infrastructure remains sovereign, auditable and resilient against geopolitical coercion or cyberattacks.
 - **Public sector modernization:** Digital public service delivery is proven to boost efficiency and user satisfaction. DPI can facilitate a ‘whole-of-government’ approach that securely enables government departments to share data, and citizens to access a wide range of services through a single interface.

- **Global collaboration:** Interoperability enables nations to share and jointly govern digital infrastructure – offering new diplomatic and cooperation opportunities amid divergent geopolitical interests and values.
- Countries without robust digital infrastructure may increasingly find themselves uncompetitive and unable to deliver essential public services. Overwhelmingly, digitalization is the preferred government approach to improve public and private services. However, adoption is not linear. As with all technology, the function and intention behind each system or platform are essential. Some applications of DPI can bring negative implications for rights and civil liberties, with risks including intrusive surveillance of citizens and the misuse of private data, particularly for the populations of authoritarian states. Therefore, transparency, democratic oversight and strong legal safeguards are vital for successful DPI that protects citizens and provides efficient public and private service delivery.

01

Introduction

Robust digital systems that can deliver efficient and reliable services are a priority for governments around the world. Digital public infrastructure offers a comprehensive route to achieving this goal.

In pursuit of efficient and reliable public services, governments across the globe have long attempted to harness the potential of digital technology. However, this agenda is no longer simply a modernization effort to cut bureaucracy and costs. It is now a strategic necessity particularly in a world where digital sovereignty, resilience, economic autonomy and alternatives to the technological dominance of the US and China are increasingly important.

Past approaches to this modernization process have often resulted in siloed and inflexible digital solutions. The digital public infrastructure (DPI) concept can facilitate a more comprehensive approach to foundational digitalization that enables the provision of modern public and private services, such as healthcare, tax collection and local authority services. Common principles – interoperability,¹ the flexibility to be updated and built upon, and the ability to securely share data – are central to successful examples of such DPI systems. The term DPI here means the principles behind this strategy as well as the resulting digital systems and platforms. A number of countries including Brazil, Estonia, India and Singapore are pioneers of robust, modern digital infrastructures that promise a two-fold benefit: a state and economy able to leverage government assets using the latest technology, and a path towards reducing the geopolitical strategic vulnerabilities and dependencies prevalent in current technology approaches.

Facing the challenges of sluggish economic growth and increasingly antiquated bureaucracies, today's decision-makers are vocal in their support of new technology solutions, referred to here as their technology stack: a collection of technical systems and software used together to implement digital applications. The emergence

¹ Interoperability means that two services, even when created or operated by different companies or organizations, can connect together. By way of example, an email client provided by Google can send emails to one provided by Microsoft or by Proton: these solutions are interoperable. By contrast, a Facebook user cannot send messages or share data between Facebook and X or YouTube. They are not interoperable.

of AI in particular has accelerated government plans for technological transitions around the world. For example, the UK's recent AI Opportunities Action plan promises to drive economic growth and modernize public services.² For many countries, the process of modernizing the state, society and the economy has taken on a new urgency in the context of a fast-changing global order, in which the pace of technology development in the US and China has the potential to expose other countries to risks related to dependency and a lack of alternatives.

Single points of technological failure and international tensions are emerging, with catastrophic, global consequences when things go wrong.

In July 2024, an alleged outage caused by a botched update to a single piece of software operated by CrowdStrike disrupted airlines, hospitals, banks and media organizations across the world, causing an estimated \$10 billion worth of damage in hours.³ Elsewhere, the Ukrainian military's reliance on Starlink satellite internet has prompted questions about operational reliability, with reported uncertainties over coverage and access during the conflict. The service operates under terms set by the US provider, which has imposed restrictions on certain uses in Crimea. This situation underscores the broader importance of diverse communication channels for government operations and civilian outreach during wartime.⁴

Public disagreements between tech giants and nation states, and a creeping sense of values misalignment, has raised the question of whether countries have sufficient sway over these vital pieces of political, cultural and information infrastructure.

The central role of giant digital platforms in search and social media has historically been approached as a regulatory challenge by countries. But public disagreements between tech giants and nation states, and a creeping sense of values misalignment, has raised the question of whether countries have sufficient sway over these vital pieces of political, cultural and information infrastructure. Speaking during the 2024 US presidential election campaign, JD Vance threatened to withdraw the US from NATO should member states continue to attempt to regulate speech on US social media platforms.⁵

² UK Department for Science, Innovation and Technology (2025), *AI Opportunities Action Plan*, <https://www.gov.uk/government/publications/ai-opportunities-action-plan/ai-opportunities-action-plan>.

³ Alegre, S. (2024), 'The CrowdStrike Outage Shows How Vulnerable We Have Become', Centre for International Governance Innovation, 21 August 2024, <https://www.cigionline.org/articles/the-crowdstrike-outage-shows-how-vulnerable-we-have-become>.

⁴ Pollet, M. (2025), 'Ukraine is stuck with Musk's Starlink for now', *Politico*, 7 April 2025, <https://www.politico.eu/article/ukraine-stuck-with-elon-musk-starlink-satellite-internet>.

⁵ Kilander, G. (2024), 'JD Vance says US could drop support for NATO if Europe tries to regulate Elon Musk's platforms', *Independent*, 17 September 2024, <https://www.independent.co.uk/news/world/americas/us-politics/jd-vance-elon-musk-x-twitter-donald-trump-b2614525.html>; Kaplan, J. (2025), 'More Speech and Fewer Mistakes', META, press release, 7 January 2025, <https://about.fb.com/news/2025/01/meta-more-speech-fewer-mistakes>.

For Western democracies, the scramble to realign with and respond to the changing priorities of the US administration under President Donald Trump has prompted a growing fear, one shared by most sovereign nations: the digital foundations on which they rely – economically, politically and culturally – are shaky and alternatives are urgently required.

Current digitalization strategies – which tend to be costly, dependent on foreign service providers, siloed, outsourced, vertically-integrated, extractive, and usually built elsewhere and hammered into shape through regulation – are unfit for a world where the sovereignty and capacity of states for technological governance are at a historically high premium. Add to this tight domestic and development budgets and drives for government efficiency, and it is clear that a new approach to digitalization is essential.

While for some in Europe this realization has been sudden – seen in the frantic activity around plans for a technological solution for digital infrastructure in the form of ‘EuroStack’, for instance – the same is not true for other countries.⁶ Alternative ways to build society-scale technology have been piloted, tested, resourced, refined and championed by leaders in India, Brazil, Kenya, Tanzania, Togo and elsewhere. A number of factors have played a part in the choices these countries made in their approaches to DPI, including concerns over high costs (as demonstrated by government IT spend in OECD countries, which is 20-times higher than it is outside of the organization);⁷ the threat of technology dependency and resulting desire for national tech champions (advocates of new technology); or the urgent need to modernize states and economies, particularly in the wake of the COVID-19 pandemic.

For countries exploring new approaches to technology governance and provision, these strategies and the technologies that underpin them – loosely grouped as DPI – can provide a path forward that is tested, modern, scalable, cooperative and immediately available. However, any DPI implementation must carefully address fundamental civil liberty concerns including data privacy, surveillance risks and the protection of anonymity. Essential safeguards include robust feedback mechanisms, government transparency and clear avenues for public redress – all critical for maintaining democratic legitimacy and public trust.

There is a difficult balance to manage. On the one hand, governments risk undermining their sovereignty by becoming overdependent on foreign technology providers. On the other hand, poorly equipped government services that fail to leverage modern technology can create and exacerbate their own governance challenges and leave citizens underserved.

When designed and implemented with strong democratic principles, DPI approaches represent both a path towards state and economic modernization and a strategic response to the technology sovereignty challenge that has emerged over the past 25 years – an issue brought so sharply into focus for Europeans since the beginning of the second Trump administration.

⁶ EuroStack (undated), ‘Building Europe’s digital future’, <https://euro-stack.eu>.

⁷ Eaves, D. (2024), ‘Lessons from digital public infrastructure’, lecture, Public AI Seminar, University College London, 20 February 2024.

02

What is DPI?

Digital public infrastructure refers to both the technological components of the digital transition as well as the underlying principles and approach. Successful DPI demands society-scale technology solutions that are interoperable, that can be built upon and that enable the sharing of data.

DPI is a comprehensive approach to building society-scale digital systems that can boost economic opportunities and enable better public services. The aim is to resolve disjointed and disparate elements of digital systems and platforms, and to improve the experience of service delivery through better efficiency and ease of use.⁸ The concept was formally recognized by the UN and the G20 in 2023, and currently informs the digitalization strategies of around 60 nations, including Argentina, Bangladesh, Colombia, Ghana, Indonesia, the Philippines, Rwanda and Saudi Arabia.⁹

DPI has emerged as an alternative to traditional approaches to building digital services, in which a vendor is contracted to provide a single, siloed solution – e.g. digital identity verification – that often cannot then be built upon, connected to other digital solutions, or be operated independent of the original vendor (Figure 1).

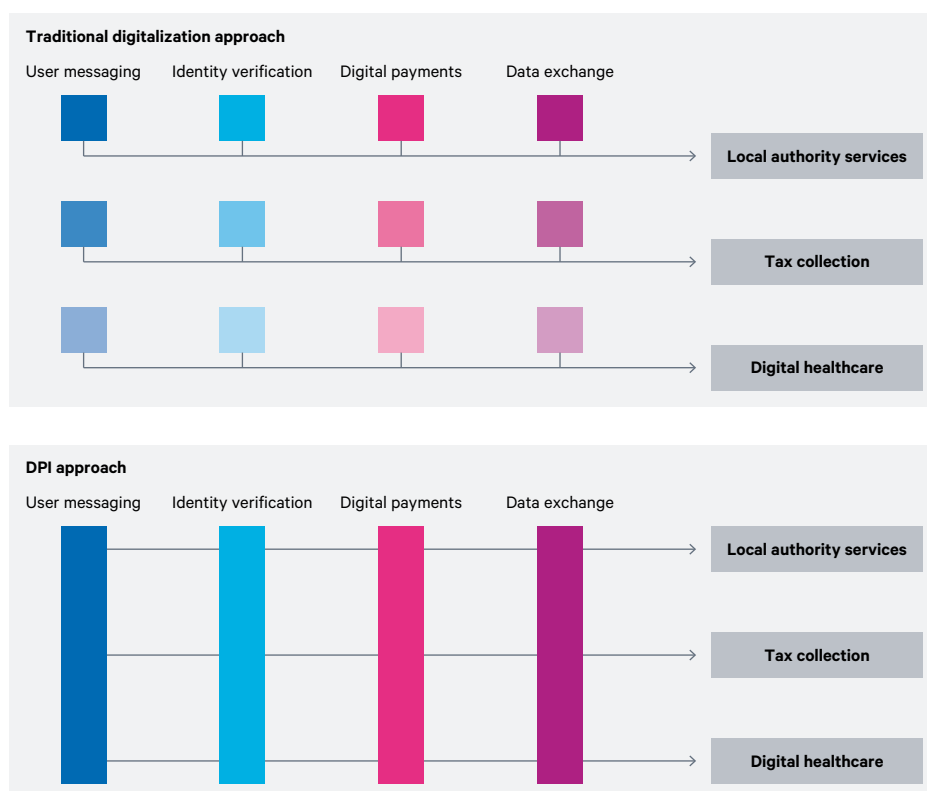
The traditional approach has led to vendors duplicating the same systems across governments around the world, though independently of one another, resulting in higher costs, siloed services and waste. The UK is replete with examples of costly duplication, vendor lock-in and inefficiency in the country's approach to digital transformation. The digitalization of the National Health Service (NHS) – a seemingly never-ending struggle in the UK – continues to be hampered by fragmentation throughout its technology ecosystems: with different electronic

⁸ Massally, K. N., Matthan, R. and Chaudhuri, R. (2023), 'What is the DPI Approach?', Carnegie Endowment for International Peace, 15 May 2023, <https://carnegieendowment.org/research/2023/05/what-is-the-dpi-approach?lang=en>.

⁹ G20, Global Partnership for Financial Inclusion (GPFI) and World Bank (2023), *G20 Policy Recommendations for Advancing Financial Inclusion and Productivity Gains through Digital Public Infrastructure: Global Partnership for Financial Inclusion*, <https://documents1.worldbank.org/curated/en/099092023121016458/pdf/P178703046f82d07c0bbc60b5e474ea7841.pdf>.

patient record systems in use across different NHS trusts, which prevents the sharing of data, trusts easily switching providers, or system expansion or updates.¹⁰ Attempts to create a single integrated NHS record that is accessible by other digital systems failed in 2011. The project was described by MPs as one of the ‘worst and most expensive contracting fiascos’ in public sector history.¹¹ Furthermore, the collapse of technology providers (e.g. UKCloud) on whom whole systems were dependent led to continuity crises and millions of pounds in underwriting, ultimately paid by the UK taxpayer.¹²

Figure 1. Traditional digitalization approach vs a DPI approach



Source: Compiled by the authors.

Note: Under the traditional digitalization approach vendors repeatedly duplicate systems leading to high costs and siloed and inconsistent services. A DPI approach provides reusable ‘building blocks’ that any digital service (public or private) can use.

In the US, COVID-19 pandemic relief was hampered by siloed state systems, some of which were nearly 50 years old, leading to ‘rampant fraud’ and delays in aid.¹³ Meanwhile, the commissioning of a healthcare portal by the state of Oregon

¹⁰ Lawrence, A., Krelle, H., Horton, T. and Gerhold, M. (2025), *Electronic patient records: why the NHS urgently needs a strategy to reap the benefits*, The Health Foundation, <https://www.health.org.uk/reports-and-analysis/analysis/electronic-patient-records-nhs-strategy>.

¹¹ BBC News (2013), ‘NHS IT system one of ‘worst fiascos ever’, say MPs’, 18 September 2013, <https://www.bbc.co.uk/news/uk-politics-24130684>.

¹² Clark, L. (2025), ‘How the collapse of local cloud provider caused biz continuity issues in UK government’, The Register, 3 March 2025, https://www.theregister.com/2025/03/03/collapse_of_ukcloud_hurt_govt.

¹³ US House of Representatives (2024), *After Action Review of the COVID-19 Pandemic: The Lessons Learned and a Path Forward*, 4 December 2024, <https://oversight.house.gov/wp-content/uploads/2024/12/12.04.2024-SSCP-FINAL-REPORT.pdf>.

encapsulates the challenge of traditional approaches to digitalization: not only did the provider fail to deliver a functioning service, but the siloed solution was only understood by that single provider, leaving the state locked into that vendor.¹⁴

The DPI approach, by contrast, requires solutions that are interoperable, that can be built upon, and that enable data to be shared.¹⁵ While this more comprehensive approach is favoured for sovereignty, resilience, security, international cooperation, modernization and economic gains, it does require large-scale data collection and storage, which poses inevitable but preventable risks to digital privacy and rights protection. To combat these risks, facilitators must support and implement strong legal and policy safeguards, privacy-enhancing technology, governance mechanisms for accountability, bias and inclusion provisions, and global standards.

India's Aadhar digital identity system is an example of the DPI approach to creating robust systems for public service provision. Rather than commissioning a single vendor to provide a digital ID solution, the Unique Identification Authority of India (UIDAI) created an open system. As a result, a range of public and private actors can complete the task of verifying a citizen's identity and providing users with a 12-digit ID number. That number can then be used for various other services, both public (e.g. tax filings, welfare payments and vaccination roll-outs) and private (e.g. the verification of ID by banks and telecoms providers, employee background checks and digital payments). Critically, this allows any future public or private services to forgo building an ID solution from scratch, piggybacking instead on Aadhar.

While initially coined to describe a handful of technologies built in this open way, the definition of the term DPI has expanded to also include global digital infrastructure like the internet and GPS. What one government or business considers a vital piece of digital infrastructure might differ from another: critical to the DPI approach is providing solutions that are open and available for others to use. Switching from siloed solutions to digital services built from interoperable blocks represents a sea change in how nation states are able to shape and exercise sovereignty over the technology on which their citizens, economies and systems of government depend.

¹⁴ Finley, K. (2014), 'What Oracle's Botched Obamacare Site Says About the Future of the Web', *Wired*, 7 March 2014, <https://www.wired.com/2014/03/oracle-oregon>.

¹⁵ Varma, P., Matthan, R., Chaudhuri, R. and Madhukar, C. V. (2024), *The Future of Digital Public Infrastructure: A Thesis for Rapid Global Adoption*, Carnegie India, https://carnegie-production-assets.s3.amazonaws.com/static/files/The_Future_of_Digital_Public_Infrastructure-_A_Thesis_for_Rapid_Global_Adoption-1.pdf.

03 Sovereignty

DPI promises greater sovereign control over digital infrastructure, reducing geopolitical dependencies and fostering self-determination.

Even countries with significant financial resources and the most developed digital sectors rely on complex international trade flows. These supply chains are characterized by monopolies, export controls and bottlenecks, particularly in cloud infrastructure and semiconductors. To a lesser extent, these restrictions also impact the availability of telecommunications equipment and critical minerals. Consequently, in provisioning technology stacks, most nations are dependent on foreign software providers, equipment and talent.

While self-sufficiency in this regard is likely impossible even for so-called ‘technology superpowers’, greater sovereignty – as measured by a country’s capacity to build and shape the technology it depends on – is eminently achievable. In recent years, in a push to strengthen sovereignty, numerous countries have reassessed both the costs and the long-term stability of their foreign technology dependencies. Singapore, Vietnam, Nigeria and Kenya have legislated that the data of their citizens must be stored in-country.¹⁶ In addition, dozens of countries have passed laws to regulate multinational digital service providers in an attempt to correct previous failures related to stewardship of their societies, markets or citizens’ lives online.¹⁷ European technology sovereignty was a core theme at the AI Action Summit in 2025, with President Macron later warning that, ‘We cannot depend on a few American companies to manage our data infrastructure, it’s a matter of European public interest... there is no such thing as happy vassalage.’¹⁸

Government innovation for public good has been at the forefront of DPI developments to date. A typical DPI approach will see governments pursuing digital projects that resolve an identifiable societal need.

¹⁶ Afina, Y. et al. (2024), *Towards a global approach to digital platform regulation: Preserving openness amid the push for internet sovereignty*, Research Paper, London: Royal Institute of International Affairs, <https://doi.org/10.55317/9781784135935>.

¹⁷ Ibid.

¹⁸ Hoesz, J. (2025), ‘Macron’s ‘Choose Europe for Science’ speech’, The French Dispatch, 5 May 2025, translation, <https://www.frenchdispatch.eu/p/macrons-choose-europe-for-science-speech-sorbonne>.

In 2016, the National Payments Corporation of India, an Indian state-owned organization, developed Unified Payments Interface (UPI), a digital system that enables instant mobile payments. This DPI, presented as a way to make the financial system more accessible to citizens, was also driven by political imperatives to establish sovereign foundations for the country's domestic digital economy. IndiaPay, one of the payment apps enabled by UPI, now serves over half a billion users, reducing transaction dependencies by processing payments in-country and tackling the dominance of monopoly players in the Indian digital payment sector.¹⁹

Sovereign ambitions also informed Estonia's X-Road project that enables and shapes the secure flow of data in the country: leaving the data of citizens more accessible to the state and government-approved businesses, and less accessible to those beyond its national borders. Given the threat posed by neighbouring Russia, Estonia is the first country to set up a 'Data Embassy', as part of a contingency plan to ensure digital continuity for its citizens. Located in Luxembourg and under heavy security, this data centre comes under Estonian jurisdiction (as would be the case for a physical embassy), through a new international law concept inspired by the Vienna Convention on Diplomatic Relations.²⁰ The X-Road platform is both secure and built on open-source technology, which enables private companies and governments in over 100 countries – such as Japan (in the energy sector), Germany (for medical prescriptions) and Brazil (in public service delivery across states) – to use the system to simplify and modernize services as well as reinforce sovereignty.²¹

Given the threat posed by neighbouring Russia, Estonia is the first country to set up a 'Data Embassy', as part of a contingency plan to ensure digital continuity for its citizens.

In another example of government-led innovation, Martha Lane Fox, the then-UK digital champion, spearheaded a project to create a digital platform to facilitate public services in the UK, which resulted in the gov.uk website in 2012. Under the UK's innovative Open Government Licence (OGL), the website has formed the basis for government-run digital services in dozens of other countries, including Canada. According to Ross Ferguson, the former director of platforms in the Canadian Digital Service, the success of the website stands out as 'an example of what is possible when governments invest in digital public goods and infrastructure'.²² Shortly after coming to power, the current UK Labour government announced its plans to bring together three digital initiatives – the Government Digital Service (GDS), Central Digital

¹⁹ UPI has itself faced scrutiny for its dominance by US-backed systems. Walmart-backed PhonePe and Google Pay together handle around 85 per cent of transactions on UPI.

²⁰ Estonia government (undated), 'Factsheet: Data Embassy', https://e-estonia.com/wp-content/uploads/factsheet_data_embassy.pdf; Sierzputowski, B. (2019), 'The Data Embassy Under Public International Law', *International & Comparative Law Quarterly*, 68(1), pp. 225–242, doi:10.1017/S0020589318000428.

²¹ X-Road (undated), 'X-Road Case Studies', <https://x-road.global/case-studies>.

²² Ferguson, R. (2023), 'The international reach of UK's Notify service', Public.digital, <https://public.digital/pd-insights/blog/2023/03/the-international-reach-of-uks-notify-service>.

and Data Office (CDDO) and Incubator for AI (i.AI) – under the Department for Science, Innovation and Technology, in order to unite efforts to improve digital public services and boost innovation.²³

The UK government’s consolidation of digital initiatives under one department is indicative of the value now placed on the development of technology solutions and the promise of emerging technology. Governments around the world have identified AI as a critical part of the global digital economy, domestic economic competitiveness and national security, as well as a force shaping politics, culture and education. There is debate over whether governments have the capacity to build or scale AI technologies when compared to private sector providers. But there is a growing realization that government-owned datasets, and data gathered from AI users, give government-backed AI technologies a unique advantage, because the data is high quality, legitimate (legally obtained) and tied to real everyday uses.

The UK’s AI Opportunities Action Plan identifies ‘Sovereign AI compute [computational power], owned and/or allocated by the public sector’ as the infrastructure that ‘will enable the UK to quickly and independently allocate compute to national priorities.’²⁴ More broadly, the Public AI Network²⁵ – an organization that promotes the use of AI for the public good – has highlighted the significant moves made by Europe, Canada, Japan, India and Singapore in pursuit of public, sovereign AI. Furthermore, through its own recent AI action plan, the US has prioritized its role in setting digital technical standards for open AI models where the country currently lags behind China, whose DeepSeek, Kimi, MiniMax and Qwen AI models all rank highly in global AI indexes.²⁶

Strengthening state sovereignty over digital technologies is not without its risks. Digital technologies are vulnerable to authoritarian abuse and overreach. For example, digital payments are more easily surveilled; single digital identities raise questions about how citizens are profiled or subjected to social scoring, such as in China’s Social Credit System;²⁷ and the collection of biometric data has obvious applications in controversial policing practices, like automated facial recognition. To date, DPI innovations have centred on technology in service of a modern, digital society, and while they have largely been built with public benefit in mind, they remain open to misuse. As such, democratic oversight, legal safeguards and meaningful public accountability are just as essential in governing DPI as they are in any other domain of public infrastructure, as was the case even before the digital age.

²³ Department for Science, Innovation and Technology, Cabinet Office and The Rt Hon Peter Kyle MP (2024), ‘DSIT bolstered to better serve the British public through science and technology’, press release, 8 July 2024, <https://www.gov.uk/government/news/dsit-bolstered-to-better-serve-the-british-public-through-science-and-technology>.

²⁴ UK Government (2025), ‘AI Opportunities Action Plan’, Department for Science, Innovation and Technology, January 2025, <https://www.gov.uk/government/publications/ai-opportunities-action-plan/ai-opportunities-action-plan>.

²⁵ Public AI Network (undated), ‘Public AI: Infrastructure for the Common Good’, www.publicai.network.

²⁶ The White House (2025), ‘Winning the Race: America’s AI Action Plan’, July 2025, <https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf>.

²⁷ Government of Canada (2025), ‘Big data and the social credit system: The security consequences’, Canadian Security Intelligence Service, July 2025, <https://www.canada.ca/en/security-intelligence-service/corporate/publications/china-and-the-age-of-strategic-rivalry/big-data-and-the-social-credit-system-the-security-consequences.html>.

There are limits to technology sovereignty that a DPI approach cannot resolve. Where dependencies run wide and deep, with entrenched incumbents and high capital investment requirements – as is the case for cloud computing, for instance – governments must temper their sovereign ambitions. While Europe’s GAIA-X cloud infrastructure is in principle an important project, it is not close to being considered a viable alternative to US providers like Amazon Web Services (AWS) or Microsoft’s Azure.²⁸ Moreover, the UK’s long-running battle with technology firms over the rollout of end-to-end encryption – such as the Apple and WhatsApp legal cases²⁹ – highlights both the limitations of regulation as a path to sovereignty and the limits that advances in technology can impose on sovereignty. Some fear that this pattern may be repeated in the future with advanced AI. Finally, ensuring digital systems are resilient to abuse and exclusion, regardless of how they have been built and deployed, remains a first-order governance challenge.

Nevertheless, faced with an unpredictable and insecure geopolitical environment, middle powers and developing countries have prioritized increased national and regional sovereignty over the technology stacks they implement. DPI approaches offer a tested route to meeting these ambitions. DPI is also a soft power mechanism that exporting countries can exercise for greater influence on the global stage. For example, had China, rather than the US, been the driving force behind the internet, the core principles of freedom of expression, open access and user privacy that today shape many digital societies of the world might have looked rather different.

Table 1. Use cases – sovereignty

Country	Issue	Existing solution*	Requirements
France	Citizens wish to avoid US or Chinese AI models.	CurrentAI – a global public interest AI partnership, aimed at reshaping the existing AI landscape by developing and supporting large-scale initiatives that serve the public interest.	Large-scale investment and government backing, strong public-private cooperation, and a clear mission.
Bangladesh	Ministers want all data on citizens to be held locally, ideally on state-owned servers.	myGov – a secure, cloud-based platform for storing and sharing verified citizen documentation and certification.	Independent building blocks such as digital ID, digital payments, trust infrastructure, or data sharing and credentials, with clear technical standards, operational guidelines and a governance structure.

* for more information see Annex 1.

²⁸ Scott, M. (2024), ‘XI. Why Europe’s Cloud Ambitions Have Failed’, in Kaltheuner, F., Saari, L., Kak, A. and Myers West, S. (eds) (2024), *Redirecting Europe’s AI Industrial Policy: From Competitiveness to Public Interest*, AI Now Institute, October 2024, <https://ainowinstitute.org/publications/xi-why-europes-cloud-ambitions-have-failed>.
²⁹ Kleinman, Z. (2025), ‘WhatsApp tells BBC it backs Apple in legal row with UK over user data’, BBC News, 11 June 2025, <https://www.bbc.co.uk/news/articles/cgmjrn42wdwo>.

04 Economic opportunity

DPI can provide an affordable path to digitalization, support domestic industries, widen access to financial tools and strengthen public finance management, such as tax collection and anti-fraud efforts.

Digitalization is a necessary condition for both modern domestic growth and participation in the global economy. However, traditional approaches to this process carry too high a cost for most countries to pursue at scale. Estimates by Professor David Eaves of University College London put the average annual spend on such projects by the governments of advanced economies at around \$250 per capita, compared with just \$14 dollars per capita in the rest of the world. To rebalance this disparity would require nearly \$1.4 trillion in new annual IT spend.³⁰ Such an investment was unrealistic even before recent tightening of government budgets. In this context, the search for new, affordable ways of building these systems should be a priority.

The DPI principles of interoperability, openness and the prioritization of public good enable countries to adopt a new, more affordable approach to building key digital systems – through the use and re-use of shared, open systems. As mentioned earlier, traditional ways of developing digital systems tend to result in expensive and inefficient silos where departments procure and maintain separate IT systems that are often locked into a single vendor for assistance. These systems are typically not interoperable with other private and public services, unable to share data with other departments, closed, or cannot be built upon for future services. To use an example from the UK, citizens are enrolled in dozens of siloed ID systems: national insurance numbers, passport numbers, NHS numbers, driving licence numbers, unique

³⁰ Eaves (2024), 'Lessons from digital public infrastructure'; Eaves, D. and Rao, K. (2024), 'What do we know about the state of DPI in the world? Preliminary insights from the DPI Map', UCL Institute for Innovation and Public Purpose, <https://medium.com/iipp-blog/what-do-we-know-about-the-state-of-dpi-in-the-world-preliminary-insights-from-the-dpi-map-51d5e49f299b>.

taxpayer references, unique pupil numbers, Proof of Age Standards Scheme (PASS) cards and so on. This expands to hundreds of IDs across the private companies and services that citizens use, from banking to online retailers, each demanding citizens prove and re-prove who they are. This duplication increases the vulnerability of users, whose data is replicated and retained in dozens of places at once, and also creates enormous inefficiencies – in contrast, Estonia claims its DPI approach to public services, enabled by a digital ID, saves more than 1,400 years of working time and 2 per cent of its GDP annually.³¹

As well as the potential budget savings, there are clear opportunities to boost economic growth. The fact that digital infrastructure underpins both the public and private sectors means government IT budgets could be spent in the service of the economy as a whole, rather than on narrow, single-use solutions.

Research from India and Brazil highlights a significant uptick in economic outcomes as cashless transfers become available and transactions become faster and cheaper.

University College London's Institute for Innovation and Public Purpose has put forward 11 ways in which the positive economic impact of a new digitalization strategy like DPI might be measured.³² These include reduced costs and increases in service quality and availability, but also capture a range of second-order market-shaping benefits: the creation of space for new businesses, increased competition through improved accessibility and expanded opportunities for consumers. Research from India and Brazil highlights a significant uptick in economic outcomes – more commercial activity, sales and higher tax revenue – as cashless transfers become available and transactions become faster and cheaper.³³ During the pandemic, countries with DPI in place before the COVID-19 outbreak were able to inject cash into their slowing economies by reaching three times more beneficiaries with emergency cash transfers than those countries without. This demonstrates the ability of DPI to provide a vital boost for the economy in times of economic strain.³⁴

DPI can give all businesses the digital tools to compete with better resourced overseas players. Mobile wallets allow customers to pay for goods and services efficiently, streamline payment processes and reduce transaction costs for business owners

³¹ Khan, I. and Shahaab, A. (2020), 'Estonia is a 'digital republic' and this may well be our future too', *Independent*, 14 October 2020, <https://www.independent.co.uk/tech/estonia-digital-republic-online-voting-blockchain-b912567.html>.

³² Eaves, D., Coyle, D., Vasconcellos, B. and Deshmukh, S. (2025), *The Economics of Shared Digital Infrastructures: A framework for assessing societal value*, IIPP Policy Report 2025/02, London: UCL Institute for Innovation and Public Purpose, <https://www.bennettschool.cam.ac.uk/publications/the-economics-of-shared-digital-infrastructures-a-framework-for-assessing-societal-value>.

³³ Dubey, T. S. and Purnanandam, A. (2023), *Can Cashless Payments Spur Economic Growth?*, SSRN, <https://ssrn.com/abstract=4373602>; ACI Worldwide(2022), 'Latin American Economies Poised to Reap Economic Benefits of Real-Time Payments as Adoption Gathers Pace – ACI Worldwide Report' 26 April 2022, press release, <https://investor.aciworldwide.com/news-releases/news-release-details/latin-american-economies-poised-reap-economic-benefits-real-time>.

³⁴ Clark, J., Marin, G., Ardic Alper, O. P. and Galicia Rabadan, G. A. (2025), *Digital Public Infrastructure and Development: A World Bank Group Approach*, Digital Transformation White Paper, Washington, DC: World Bank, <https://openknowledge.worldbank.org/entities/publication/cca2963e-27bf-4dbb-aa5a-24a0ffc92ed9>.

by removing third parties. Pix, an instant payment platform created and managed by the Central Bank of Brazil, reduced transaction costs for its users – with an average cost to merchants of 0.22 per cent, compared with fees of 2.2 per cent for credit cards in Brazil.³⁵ The reduced costs for its 153 million users both support Brazilian small and medium-sized enterprises and contribute to a more balanced, inclusive and dynamic marketplace.³⁶

Alongside direct benefits, payment systems made possible through DPI shape the economy by increasing market entry points and enabling new groups to participate more fully in financial systems. With the introduction of Aadhar in India, the cost of verifying a customer's identity plummeted from \$10–\$12 to about \$0.20, making it feasible to serve millions in low-income and remote settings and enabling economically disempowered people to save, invest, access credit and better control their finances.³⁷ According to the World Bank, the number of Indian adults with a bank account more than doubled between 2011 and 2021, which studies attribute to cheaper and more widely-available digital ID checks and digital payment systems.³⁸ Between 2006 and 2013, Kenya's dominant mobile payment system, M-PESA, contributed to the country's 3.4 per cent per capita real income growth.³⁹ As sign-ups to the platform increased, so too did economic gains.

There is evidence that DPI approaches present further opportunities for improved public finance management. In Kenya, 59 per cent of the country's GDP flows through M-PESA. This has drawn firms operating in the informal sector into the tax net and provided new opportunities for collecting tax. In 2024, the Kenyan government imposed a controversial 3 per cent duty on M-PESA transactions, which critics say counters the benefits of financial inclusion.⁴⁰ DPI that brings together identity systems and direct payments can reduce instances of fraud: India's government reported its Direct Benefit Transfer system enabled quick distribution of COVID-19 relief funds, 'ensuring accurate targeting of the beneficiaries, de-duplication and reduction of fraud'.⁴¹ By contrast, many advanced economies reported rampant fraud in the distribution of COVID-19 relief funds. An Associated Press analysis estimates \$280 billion of US COVID-19 relief funding was lost to fraud.⁴²

³⁵ Duarte, A. et al. (2022), *Central banks, the monetary system and public payment infrastructures: lessons from Brazil's Pix*, Bank of International Settlements (BIS), No 52, <https://www.bis.org/publ/bisbull52.pdf>.

³⁶ Jain, S. (2025), 'What happens to an economy with digital public infrastructure?', Gates Foundation, 16 April 2025, <https://www.gatesfoundation.org/ideas/articles/inclusive-economies-digital-public-infrastructure#:~:text=2.,banks%2C%20cooperatives%2C%20and%20fintechs>.

³⁷ Ibid.

³⁸ Klapper, L., Singer, D. and Ansar, S. (2021), 'The Global Findex Database 2021, India Country Brief', World Bank, <https://thedocs.worldbank.org/en/doc/0e271c416d4afa3237cf3a6226884d3b-0430062023/original/Findex2021-IndiaCountryBrief-DRAFT-2-03-23.pdf>.

³⁹ World Economic Forum (2015), 'How mobile money is driving economic growth', 15 September 2015, <https://www.weforum.org/stories/2015/09/how-mobile-money-is-driving-economic-growth>.

⁴⁰ ITEdgeNews (2024), 'M-Pesa 2.0: The case for reimagining Kenya's payment infrastructure', 24 February 2025, <https://www.itedgenews.africa/m-pesa-2-0-the-case-for-reimagining-kenyas-payment-infrastructure>.

⁴¹ Government of India, Ministry of Road Transport and Highways (2024), 'JAM (Jan Dhan, Aadhar, Mobile) TRINITY and digital revolution: A Decade of Financial Inclusion, Transparency and Corruption Free India', press release, 20 December 2024, <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2086611>.

⁴² Lardner, R., McDermott, J. and Kessler, A. (2023), 'The Great Grift: How billions in COVID-19 relief aid was stolen or wasted', AP News, 12 June 2023, <https://apnews.com/article/pandemic-fraud-waste-billions-small-business-labor-fb1d9a9eb24857efbe4611344311ae78>.

Replicating outdated methods of digitalization is not feasible for most nations. Using DPI principles can provide a credible and cost-effective digitalization strategy. With relatively low implementation costs compared to standard digitalization approaches, DPI strategies can utilize a range of different financial resources, such as philanthropic funding or political union support. MOSIP (Modular Open Source Identity Platform), a digital ID system with over 100 million users worldwide, was launched through \$28 million of philanthropic support from the World Bank, Gates Foundation and the Omidyar Network, among others. Today, MOSIP has an estimated annual operating budget of between \$5 million and \$7 million, yet the project has engaged over 25 countries across Asia Pacific, Africa, Latin America and the Caribbean.⁴³ Similarly, a 2018 extension of Estonia's X-Road to enable data sharing across European borders, with an estimated cost of €252,000, had full EU support through the European Regional Development Fund, which offered to cover up to €352,000 through the Operational Programme for Cohesion Policy Funding for the 2014–2020 period.⁴⁴ These projects demonstrate high engagement, cost-effectiveness and investment diversification.

By circumventing the mistakes of siloed digital services, a DPI approach lays a strong foundational layer for a country's economy, reducing costs, contributing to growth and unlocking new avenues for financial inclusion, participation, innovation and efficiency across sectors.

Table 2. Use cases – economic opportunity

Country	Issue	Existing solution*	Requirements
Togo	During the COVID-19 pandemic, citizens struggled to feed their families.	NOVISSI – an emergency cash transfer programme that supported informal workers whose incomes were disrupted. The programme was later extended to rural areas with higher poverty incidence.	Mobile phone access, satellite imagery, well-maintained demographic data and strong government leadership.
India	Small-business owners are frustrated that card-based payment systems have increased the costs of using their services.	Unified Payment Interface (UPI) – an instant real-time payment system that enables instant money transfers between bank accounts using a mobile device.	A supportive banking system, access to mobile phones and regulator alignment across authorities.
Kenya	There is often no physical bank branch in smaller towns and rural communities, so access to bank accounts is limited for many people.	M-PESA – a system that enables users to send, receive and store money using mobile phones. The system opens economic entry points for people otherwise cut off from financial systems.	Strategic private partnerships, secure telecom infrastructure, regulatory flexibility and user-centred design.

* for more information see Annex 1.

⁴³ Hariharan, V. (2024), *Lessons Learned: Reflecting on MOSIP's Journey to Scale*, Digital Impact Alliance, <https://dial.global/wp-content/uploads/2024/03/MOSIP-Lessons-Learned.pdf>.

⁴⁴ European Commission (2018), 'X-Road – cross-border co-development of national data exchange platform', https://ec.europa.eu/regional_policy/en/projects/europe/x-road-cross-border-co-development-of-national-data-exchange-platform.

05 Resilience and security

Through its adaptability and openness, DPI boosts resilience by reducing single points of failure and dependencies that can be weaponized by adversaries.

In the national security context, DPI can bolster resilience and safeguards through enabling adaptability and mitigating systemic risks.

Calls to remove Russian banks from the SWIFT payment system (a significant global financial player) following Russia's invasion of Ukraine were initially met with alarm by some states in the EU, as the situation brought into focus their own vulnerabilities. From their perspective this sanction measure was a gamble: with Germany, France and Italy concerned that companies owed money by Russian businesses would struggle to be paid, and consequently their economies would suffer. It also brought to light the fact that if SWIFT could be used against a G20 economy, it could theoretically be used against any country that falls foul of powerful states that control access to the payment system.

The dependence on digital systems located in other countries that are predominantly controlled and influenced by both powerful state actors and multinational providers is a growing concern. A DPI approach requires an evaluation of whether existing dependencies are secure and adaptable to changing geopolitical conditions, an assessment of the costs for mitigating any identified risks and a blueprint for addressing those vulnerabilities.

Using DPI that can be controlled domestically, countries can create sovereign alternatives in case of global digital disruption or political instability. Robust domestic digital payment infrastructure, for instance, can support a state's internal economy in the short term, even if access to international systems is disrupted.⁴⁵

⁴⁵ Khiaonarong, T., Leinonen, H. and Rizaldy, R. (2021), 'Operational Resilience in Digital Payments: Experiences and Issues', *International Monetary Fund*, 2021(288), <https://www.elibrary.imf.org/view/journals/001/2021/288/article-A001-en.xml?ArticleTabs=fulltext>.

DPI typically positions the governance, data and operational control of critical digital systems within national borders or under publicly accountable entities. This contrasts sharply with pursuing seemingly cheaper or more convenient siloed technology solutions by relying on foreign-based corporations or international bodies influenced by a few powerful states. Sovereign control can enhance a country's ability to ensure the system serves national interests, is flexible to change and cannot be easily shut down or manipulated externally.

Innovation in domestic digital infrastructure capacity has been felt most keenly in conflict and crisis. The Russian invasion of Ukraine significantly accelerated Kyiv's pursuit of resilient DPI solutions to public service provision, originally a 2019 campaign pledge by President Volodymyr Zelenskyy. Ukraine's sovereign Diia platform – launched just before, and perhaps in anticipation of, the invasion – was designed to federate data storage and be resilient to cyberattacks, and it was adapted quickly to provide new services in war, including the distribution of war bonds and secure channels to report Russian troop movement.⁴⁶ It successfully maintained access to citizen services after February 2022 by migrating to distributed cloud servers with the support of US providers.⁴⁷ However, Ukraine's dependence on foreign – predominantly US-based – cloud and cybersecurity service providers still raises complex questions about national resilience and DPI.

Elsewhere, digital ID and direct digital payment systems are credited with recently enabling the Bangladeshi government to pre-emptively distribute anticipatory mobile transfers of \$37 to each person likely to be affected by forecasted floods, improving readiness and reducing the impact of floodwaters.⁴⁸

DPI approaches emphasize open standards, interoperable solutions and modular components. These principles contribute to a country's digital resilience by reducing single points of failure: should a system fail, the impact is contained, and alternative or complementary systems can continue to function, ensuring adaptability and continuity of essential services and faster recovery. Moving away from siloed solutions broadens familiarity with common digital systems, thereby enhancing internal expertise, organizational capacity and resilience.

The participation of multiple vendors and operators in the development and supply of a digital system further reduces the risks associated with being reliant on one vendor. As previously mentioned, the 2024 CrowdStrike glitch that crippled hospital systems and grounded air traffic following a faulty software update underscored the vulnerability of critical systems to single points of failure.⁴⁹ Despite prompting calls for urgent reviews of critical digital systems, single-vendor provision of operating systems, cloud computing and network infrastructure remains the default. Commenting on a review by the US Cyber Safety Review Board into a major government security breach, Senator Ron Wyden stated that

⁴⁶ Lawler, D. (2023), 'Ukrainians use the same app to file taxes and track Russian troops', AXIOS, 26 May 2023, <https://www.axios.com/2023/05/26/ukraine-diia-app-fedorov-russian-troops>.

⁴⁷ New America (2023), 'Digital Public Infrastructure in Ukraine: Harnessing Technology for the Public Good', <https://www.newamerica.org/the-thread/digital-public-infrastructure-ukraine-war>.

⁴⁸ United Nations Development Programme (2024), 'From Response to Resilience: How Digital Financial Services Can Advance Climate Adaptation Efforts Better Than Cash Alliance contribution', <https://sdgfinance.undp.org/news-events/response-resilience-how-digital-financial-services-can-advance-climate-adaptation>.

⁴⁹ Privacy International (2024), 'CrowdStrike: What the 2024 outage reveals about security', <https://privacyinternational.org/long-read/5507/crowdstrike-what-2024-outage-reveals-about-security>.

although ‘the government’s dependence on Microsoft poses a serious national security threat’, given the federal government’s overwhelming dependence on the technology provider it would be very difficult to replace the company.⁵⁰ Similarly, there are growing European fears about the continent’s dependence on Microsoft products, with an internal European Commission review judging that ‘there are no known credible offerings from European providers’ that could replace the US software giant.⁵¹

While a DPI approach boosts security it also has second-order benefits for national resilience and productivity. By committing to developing and maintaining resilient digital infrastructure, a nation must foster a robust pool of its own digital talent. This growth in domestic expertise reduces reliance on foreign specialists for critical maintenance, ongoing development and rapid responses to cybersecurity threats. In addition, it ensures that the knowledge and skills necessary to secure and steward a nation’s digital foundations can be found at home. The UK’s AI strategy explicitly names public and private UK talent as a central pillar.⁵² For organizations such as NATO, improvements in domestic digital capacity are central for extending and bolstering intergovernmental resilience.

By taking responsibility for the provision of robust digital systems, Ukraine and other countries have shown a credible path forward for digital infrastructure that strengthens national resilience and security.

Table 3. Use cases – resilience and security

Country	Issue	Existing solution*	Requirements
Malawi	It is hard for the government to reliably verify citizen identity. This is having an impact on border control, and by extension, counterterrorism and immigration policy.	National Registration System – a permanent and continuous national registration and identification system, including a physical ID card that is accessible and verifiable on smartphones.	Reform of the National Registration Act, investment and development in hybrid ID systems (including from UNDP) and capacity-building.
Ukraine	The government is having to refocus its resources following the full-scale invasion, resulting in reduced access to government documents and public services.	Diia – a platform to federate data storage, improve cybersecurity and provide services and citizen support during the conflict. The system allows users to report the position of Russian soldiers, take part in opinion polls, process tax payments and filing, access e-services, make army donations, and access e-restoration – where citizens can obtain compensation for destroyed or damaged housing as a result of Russian aggression.	Interoperable systems across departments, complete and secure citizen databases, robust cybersecurity standards and highly skilled internal technical expertise.

* for more information see Annex 1.

⁵⁰ The Register (2024), ‘US government excoriates Microsoft for ‘avoidable errors’ but keeps paying for its products’, 5 April 2024, https://www.theregister.com/2024/04/05/microsoft_government_contracts.

⁵¹ Wulff Wold, J. (2025), ‘Internal documents reveal Commission fears over Microsoft dependency’, Euractiv, 9 January 2025, <https://www.euractiv.com/section/tech/news/internal-documents-reveal-commission-fears-over-microsoft-dependency>. In April 2025, Microsoft announced a set of ‘European digital commitments’, noting that ‘in a time of geopolitical volatility, we are committed to providing digital stability’. Commitments included expanding cloud and AI infrastructure, upholding Europe’s digital resilience and continuing to protect and defend Europe’s cybersecurity. Smith, B. (2025), ‘Microsoft announces new European digital commitments’, Microsoft press release, <https://blogs.microsoft.com/on-the-issues/2025/04/30/european-digital-commitments>.

⁵² UK Government (2025), ‘AI Opportunities Action Plan’.

06

Public sector modernization

DPI can transform bureaucratic efficiency and service delivery through automation and interoperability.

The rapid pace of digital innovation, from generative AI to autonomous vehicles, is often attributed to the private sector. However, this narrow view misses a crucial element: governments are also significant innovators in their own right. Understanding and supporting public sector innovation is essential for ensuring that technological progress serves the public good and addresses societal needs.

Public sector modernization through DPI involves integrating otherwise siloed solutions into unified, foundational digital infrastructure. As an illustration, while UK citizens can access a range of public services online as varied as financial support, voter registration, birth certification, student loans or even pet registration, each requires engagement with an individual digital system. By contrast, a UK Google user has access to a myriad of Google-owned and third-party applications through a single login and interface. The ambition of DPI is to make the former more like the latter.

But simply rolling out a digitalization agenda will not, on its own, transform service delivery. First, equitable access to technology is essential for enabling efficient, inclusive and transparent public service delivery. Without addressing the digital divide, DPI or any digitalization strategy will not maximize the benefits of technology. Second, in addition to sustained funding and direction, meaningful change requires public transparency, citizen support and oversight to protect users against the misuse of their data and enable the deep reform of underlying bureaucratic systems. In successful cases, DPI is accompanied by a fundamental rethinking of how public services are conceived, built and delivered – starting with the perspective of users, rather than that of government agencies or technology providers. This requires states to adopt a human-centred method for improving public services through digitalization. This means mapping all in-person and

digital touchpoints in delivery and designing cohesive end-to-end experiences, pursuing modernization not as a goal in itself, but as a means to achieve more accessible, efficient and equitable outcomes for all citizens.

At a minimum, DPI systems provide convenience for the public and cost-efficiency for the government. But, for the more ambitious, DPI presents an opportunity to facilitate a stronger and more secure social contract, where in return for basic data and identity details, consenting citizens receive timely, beneficial and practical information from the government, greater satisfaction in service delivery and reaffirmed – or re-established – trust in government. Like Ukraine's Diia, Estonia's X-Road is described by the Estonian government as 'the backbone' of its digital state, providing access to over 3,000 public and private services and facilitating the sharing of data between those services and agencies efficiently and securely.⁵³ The open-source code of the X-Road system has been shared with other countries. In Singapore, the national digital ID, Singpass, also promises 'secure and convenient access to over 2,700 services across 800 government agencies and businesses'.⁵⁴

Table 4. Use cases – public sector modernization

Country	Issue	Existing solution*	Requirements
UK	Recipients of disability benefit must travel to a government building to receive payment. This can be extremely difficult in terms of accessibility and is an inefficient use of their time.	gov.uk – the UK government's official website, providing a central hub for government services and information.	Political and financial support for digital solutions and a higher risk tolerance for mistakes along the way.
Estonia	There are prevalent inefficiencies and duplicated efforts across government agencies. Public services would be far better if departments could access databases across government.	X-Road – open-source software that provides unified and secure data exchange between organizations in a collaborative ecosystem.	Political support of open-source software and ecosystem solutions and a secure and scalable data exchange platform.
Brazil	Filing and processing divergent forms to meet basic public service provisions is not only unsatisfying but makes the country feel stuck in the past.	Identificação Civil Nacional – provides each citizen with a national unique identity number that will carry other official information from other governmental identity documents.	Full system government digitalization strategy, centralization of databases, data transfers between public administration bodies, and a review of data protection legislation.

* for more information see Annex 1.

The introduction of digital governance systems, or e-governance, is often the first step in developing DPI. In 2024, Estonia's capital city Tallinn, Paris and Riyadh were rated in the top three spots of the UN's Local Online Services Index, due to their provision of highly proficient digital institutional frameworks that both facilitate the delivery of information and public services and boost participation

⁵³ e-estonia (2025), 'X-road – Interoperability services', <https://e-estonia.com/solutions/interoperability-services/x-road>.

⁵⁴ GovTech Singapore (2025), 'Singpass', <https://www.tech.gov.sg/products-and-services/for-citizens/digital-services/singpass>.

and engagement, e-government literacy and technology.⁵⁵ In Denmark, the country's Digital Post – a cross-government digital postal service infrastructure – enables public authorities to communicate securely and digitally with citizens and businesses (and vice versa). By digitizing communication and increasing interoperability with external private senders, such as banks and utility services, the postal system further saves the Danish government €270 million annually.⁵⁶ Meanwhile, in South Korea, 'smart cities' were developed to provide technology solutions to transport, public security, disaster prevention, administration, healthcare and childcare. The government of Busan predicted that, in 2022, these initiatives would save the city's residents up to 124 hours a year, including 60 hours spent on daily commutes, 20 hours on administrative work and 5 hours waiting at hospitals.⁵⁷

While authorities often establish in-house departments for setting, leading and delivering digital modernization in government, in some areas, public-private partnerships help to plug the gaps in investment, R&D and innovation.

In finance modernization, 'open banking' has paved the way for a more transparent digital economy by providing secure data sharing between banks and third-party organizations, like fintech companies. For those with reduced access to physical bank branches, digital payment platforms such as Stripe, Plaid, Truelayer and UPI provide an improved and updated entry point to finance and public services. In the UK, this includes citizens' ability to use open banking for HMRC tax payments and refunds, streamlining payments, reducing costs, cutting bureaucracy and improving fraud prevention.⁵⁸ However, policymakers should be wary of commercial software or technology consultation masquerading as DPI, with vendors marketing their proprietary tools as the only solution to complex infrastructural problems. For example, the NHS Federated Data Platform – a system designed to create a common data infrastructure across the NHS – might enhance patient care, but the software provided is not a digital public good – it was not built with transparency, collaboration, or open principles in mind.

Public services need to benefit citizens, work for government coffers and strengthen trust in public institutions. Modern and transformative DPI needs to utilize digital technology effectively, but also be reusable, interoperable and built on top of existing architecture. Partnerships, between government departments and with the private sector, can plug gaps in interoperability and help to prioritize accessibility, renewal and a unified user experience across government touchpoints.

⁵⁵ UN DESA (2024), 'Local E-Government Development', in UN DESA (2024), *E-Government Survey 2024*, <https://desapublications.un.org/sites/default/files/publications/2024-09/%28Chapter%204%29%20E-Government%20Survey%202024%201392024.pdf>.

⁵⁶ European Commission (2013), 'Denmark: Digital mailboxes in all public institutions and companies', Shaping Europe's digital future, news article, 22 November 2013, <https://digital-strategy.ec.europa.eu/en/news/denmark-digital-mailboxes-all-public-institutions-and-companies>.

⁵⁷ Teh, N. and Martina, A. (2019), *Global Expert Mission South Korea Smart Cities 2019*, UKRI and Global Alliance, https://iuk-business-connect.org.uk/wp-content/uploads/2020/12/B2_15_KTN_SouthKorea_SmartCities.pdf.

⁵⁸ HM Revenue & Customs (2024), 'Open Banking privacy notice', 22 May 2024, <https://www.gov.uk/government/publications/open-banking-privacy-notice/open-banking-privacy-notice#:~:text=Open%20Banking%20allows%20you%20to,in%20to%20your%20HMRC%20account>.

07 Global DPI collaboration

DPI enables global collaboration on shared challenges without countries having to surrender autonomy. It can also act as a secondary channel for diplomacy.

As advocated by the UN's Global Digital Compact, international collaboration on and through DPI involves interoperable systems that respect sovereignty while promoting shared resources and networks.⁵⁹ These digital systems create and reinforce collaborative models, capacity-building and standard-setting. As a result, this digital infrastructure offers a new avenue for cooperation and diplomacy in an ever-fragmented global order.

Interoperable systems are often built using open-source software (OSS), where the source code is publicly available and can be used, modified and distributed freely by anyone and for any purpose. OSS licences grant users, including governments, these rights, with the aim of fostering greater collaboration and digitalization. These shared systems can lower costs, increase the flexibility of application and enhance security and reliability through community scrutiny.⁶⁰

The COVID-19 pandemic underscored the importance of OSS with the mutual recognition of digital vaccination certificates. In partnership, the World Health Organization (WHO) and EU created the EU Digital COVID Certificate, adopted by all 27 EU member states. This system used open-source code and common principles to share healthcare information, which led to reduced transmission and saved lives. This approach enabled EU countries to leverage digital functions for health, while maintaining national guidelines on pandemic responses.⁶¹

⁵⁹ United Nations General Assembly Resolution (2025), *Pact for the Future, Global Digital Compact and Declaration on Future Generations*, https://www.un.org/sites/un2.un.org/files/sotf-pact_for_the_future_adopted.pdf.

⁶⁰ Open Source Initiative (2025), 'Growing the impact of Open Source around the world', <https://opensource.org>.

⁶¹ European Commission (2023), 'EU Digital COVID Certificate', https://commission.europa.eu/strategy-and-policy/coronavirus-response/safe-covid-19-vaccines-europeans/eu-digital-covid-certificate_en.

Effective examples of open digital infrastructure during the COVID-19 pandemic accelerated global acceptance of interoperable systems, noticeably shifting the balance of interoperable digital infrastructure from intra-jurisdictional to cross-border. As a result, WHO, in collaboration with Saudi Arabia, has developed the Hajj health card scheme to support 3 million people travelling from Indonesia, Malaysia and Oman to Mecca for the annual pilgrimage.⁶² This open-source platform, built on robust and transparent standards, improves the safety of international pilgrims by providing a secure way for participating states to share personal data with healthcare providers, such as medication needs, allergies, immunization status and pre-existing conditions. This DPI tool increases data security and trust among participating countries.

Effective examples of open digital infrastructure during the COVID-19 pandemic accelerated global acceptance of interoperable systems, noticeably shifting the balance of interoperable digital infrastructure from intra-jurisdictional to cross-border.

International collaboration through DPI is not confined to healthcare. In Singapore, the Infocomm Media Development Authority created TradeTrust, a successful digital framework for trade that allows the exchange and verification of digitized documents. For electronic bills of lading for routes between Singapore and Rotterdam, TradeTrust has reduced the processing time from 5–10 days to under 24 hours.⁶³ This has further cemented Singapore's position as a secure, efficient trading partner, and a fintech leader in Southeast Asia. In the EU, the electronic Identification, Authentication and trust Services (eIDAS) framework, a comparable system for ensuring the authenticity of electronic signatures and documents, has achieved similar outcomes for improving international trust and trade.

There is insufficient funding for open-source DPI at present, which diminishes collaborative and digital diplomacy gains. The Digital Public Goods (DPG) Registry – a global multi-stakeholder initiative – is an exemplary record of digital tools that have open software, standards, data, AI systems and content that adhere to privacy standards and the attainment of the UN Sustainable Development Goals (SDGs).⁶⁴ Likewise, the DPI Use Cases Explorer, offers a repository of applied DPI that deliver real-world services.⁶⁵ But these registries are not exhaustive, and gaps remain. Without easily accessed interoperable systems, states will look for alternative sources for DPI. In these instances, developing and emerging countries are often co-opted into adopting either Global North or Chinese technological standards and systems. This problem is demonstrated by Pakistan's reliance on Beijing's China–Pakistan

⁶² World Health Organization (2024), 'WHO global network expands digital health certification for Hajj pilgrims', <https://www.who.int/news/item/21-10-2024-who-global-network-expands-digital-health-certification-for-hajj-pilgrims#:~:text=The%20Hajj%20health%20card%2C%20built,status%20and%20pre%2Dexisting%20conditions>.

⁶³ Yong, S. L. (2022), 'Trade Digitalisation TradeTrust', World Trade Organization, https://www.wto.org/english/res_e/reser_e/pres_loh.pdf.

⁶⁴ Digital Public Goods Registry (2025), 'DPG Registry', <https://www.digitalpublicgoods.net/registry>.

⁶⁵ DPI Use Cases Explorer (2025), 'DPI Use Cases Explorer', <https://dpiexplorer.org>.

Economic Corridor digital systems, exposing critical infrastructure to debt traps;⁶⁶ African nations adopting Eurocentric General Data Protection Regulation (GDPR) standards due to colonial legal ties;⁶⁷ and in the pursuit of AI capabilities, where middle powers and developing countries are presented with a binary choice between US and Chinese technology.⁶⁸ Addressing this power imbalance is a critical issue.

Greater awareness of and funding for OSS would provide countries with more options, as would new interoperable systems, more cooperation, digital exchange and improved practice between states using similar DPI systems.

To date, a specific framework for DPI cooperation has not been established. But organic, effective collaboration on shared challenges is already happening through digital means. A formalized collaborative governance mechanism would be the next step to protect interoperable systems from potential disputes due to differing legal systems, while respecting autonomy and different standards. The final form of such a mechanism is yet to emerge, but lessons could be learned from Gavi, the Vaccine Alliance, on public–private partnership, co-financing and making the most of a lean governance structure for strategic direction and policymaking.

Table 5. Use cases – global DPI collaboration

Country	Issue	Existing solution*	Requirements
European Union (EU)	Citizen digital health records are not valid when they travel to other EU member states.	EU Digital COVID Certificate (built on the Global Digital Health Certification Network) – brought about greater alignment on standards and accreditation for digital vaccination certificates, and therefore improved health policies that facilitated international and domestic travel.	Open standards and software, regional political support and well-maintained health datasets.
Singapore	Issuing a bill of lading between a buyer in Italy and the shipping company in Singapore takes too long and disincentivizes engagement and trade partnerships.	TradeTrust – comprises globally accepted standards that connect governments and businesses to a public blockchain to enable trusted interoperability of electronic trade documents across digital platforms.	Strong public–private partnership and globally accepted standards for accreditation.
Saudi Arabia	Performing Hajj is a sacred pilgrimage, but many are worried that they will become unwell while travelling and do not complete this important religious duty.	The Hajj health card (built on the Global Digital Health Certification Network) – increased medical data sharing for better healthcare communication and delivery between nations engaged in Hajj.	Internationally shared goals and cultural traits, sharable health datasets and open standards and software.

* for more information see Annex 1.

⁶⁶ Afzal, M. (2020), “At all costs”: How Pakistan and China control the narrative on the China-Pakistan Economic Corridor, The Brookings Institute, June 2020, <https://www.brookings.edu/articles/at-all-costs-how-pakistan-and-china-control-the-narrative-on-the-china-pakistan-economic-corridor>.

⁶⁷ Bryant, J. (2021), ‘Africa in the Information Age: Challenges, Opportunities, and Strategies for Data Protection and Digital Rights’, *Stanford Technology Law Review*, 24(2), pp. 389–439.

⁶⁸ Krasodonski, A. (2025), ‘The US–China AI race is forcing countries to reconsider who owns their digital infrastructure’, Chatham House Expert Comment, 12 May 2025, <https://www.chathamhouse.org/2025/05/us-china-ai-race-forcing-countries-reconsider-who-owns-their-digital-infrastructure>.

08

Conclusion and recommendations

DPI can be transformative for societies, but it requires commitment to shared standards, steadfast implementation and strategic investment. This approach can be particularly challenging in a context of tight government budgets and shortsighted leadership.

This research paper has presented the case for adopting a DPI approach that enables governments to develop more sovereign, secure and modern social technology ecosystems. It has outlined the value that DPI can bring to sovereignty, resilience, security and public sector modernization, in addition to the economic opportunities and the role of DPI in promoting global collaboration. Throughout, the paper has highlighted important elements of the increased global uptake of DPI approaches, from Brazil to Ukraine. The analysis here is bullish on the need for new digitalization strategies, while recognizing that the more comprehensive migration of our lives online demands appropriate governance to prevent the abuse of new forms and concentrations of power.

DPI is more than a technical solution: it is a new theory of change that reframes the digital state as a platform for innovation and sovereignty. Countries are looking to renegotiate and reinvigorate the stewardship of their societies' digital foundations as a strategic necessity in a world of geopolitical instability, technology sovereignty struggles and tight public budgets.

There is a growing research and evidence base on how countries might embrace a DPI approach, from the G20 declarations and World Bank programmes to work by global philanthropic and research organizations.⁶⁹ States must begin with a set of guiding principles – interoperability, reusability, openness and the ability to securely share data – that shape both the design and the governance of digital infrastructure. The lessons from DPI show that to successfully revolutionize a country's digital infrastructure, these principles must be at the centre of any DPI or sovereign technology strategy.

States must begin with a set of guiding principles – interoperability, reusability, openness and the ability to securely share data – that shape both the design and the governance of digital infrastructure.

- First, governments must continue to strengthen their capacity to shape technology beyond regulation and rule-making. From the US to China, and in middle powers like the UK and France, new investment – for example in attracting talent, digital literacy and civil service up-skilling – is strengthening state capacity to effectively procure, maintain and design digital public services.
- Second, governments can boost interoperability across the public and private sectors by committing to openness. When government technology (such as gov.uk) is designed using a range of open standards and safeguards – such as open application programming interfaces (APIs), open documentation, open licensing and open-source codebases (the collection of source code used to build software) – it creates a robust and flexible foundational layer of technology that others can develop and build upon. This ensures interoperability, reduces vendor lock-in and supports future adaptation without costly reinvention.
- Lastly, collaboration is strength. No state should expect to build digital technology in a vacuum. Partnerships (regional, multilateral, mini-lateral, public–private and technical) increase resilience, accelerate learning, reduce costs and create shared leverage in shaping global norms.

⁶⁹ Government of India (2024), 'Declaration on Digital Public Infrastructure, AI and Data for Governance – Joint Communiqué by the G20 Troika (India, Brazil and South Africa), endorsed by several G20 countries, guest countries and international organizations', 20 November 2024, <https://www.mea.gov.in/bilateral-documents.htm?dtl/38551/Declaration+on+Digital+Public+Infrastructure+AI+and+Data+for+Governance++Joint+Communiqu233+by+the+G20+Troika+India+Brazil+and+South+Africa+endorsed+by+several+G20+countries+guest+countries+and+international+organizations>; World Bank Group (2025), 'Digital Public Infrastructure and Development: A World Bank Group Approach – Digital Transformation White Paper: Volume 1', 31 January 2025; Gates Foundation (2025), 'Digital Public Infrastructure', <https://www.gatesfoundation.org/our-work/programs/global-growth-and-opportunity/digital-public-infrastructure>.

Recommendations

Moving from principles to practice requires coordinated action across multiple domains. There is no single blueprint, but there are common steps that can guide implementation.

1. Establish a political and institutional mandate for DPI

A DPI approach begins with a clear political commitment that recognizes digital infrastructure as a strategic asset and as a critical element of public sector reform. In many cases this has been done through intergovernmental bodies or forums, but a more concerted political effort is also needed to prioritize and make the case for digital infrastructure at the national and local levels.

Alongside political leadership, institutional mechanisms are essential to encourage citizens and allay any concerns over the adoption of this technology. This could take the form of mandated government departments like the Department for Science, Innovation and Technology in the UK; US-style public-private taskforces, where government agencies and private entities collaborate to deliver and finance public services; or a digital architecture board, with statutory powers that promote the adoption of common standards and reusable solutions to drive technology alignment across government. This is particularly important in political environments where the risk appetite for government innovation is low. This can include countries with heavily embedded bureaucratic legacy structures, governments interested in short-term gains, states with powerful interest groups, and those with a public that values tradition over innovation and where citizens show concern or distrust about the ability of governments to protect digital privacy.

2. Grow technical capacity

Modern states must develop public and private institutions capable of designing, stewarding, operating, procuring and regulating those technologies identified as critical to a nation's digital sovereignty. Middle power and developing countries must conduct comprehensive and realistic assessments of their achievable domestic capabilities – as opposed to carrying out short-term box-ticking exercises – and pursue complementary partnerships that are both strategic and mutually reinforcing.

Capacity must grow within the public sector – if necessary, using investment from the open-source community and private actors. As stated earlier, financing DPI is more cost-effective in the long-term than existing investment methods for unfit, siloed digital schemes. In addition, states must also take steps to reform and modernize government workplace culture and collaboration, stabilize institutions, implement transparent procurement procedures, and make strategic investments in infrastructure, technology and, crucially, human capital. Governments should look to recruit and retain technical talent with strategic visa policy, upskilling opportunities and competitive salaries.

Useful models for increasing technical capacity include the UK's Advanced Research and Invention Agency (ARIA) and Government Digital Services (GDS), both of which prioritize operational agility, innovative 'sandboxing' and independence. Another possibility is the creation of publicly mandated focused research organizations

(FROs), successful examples of which include the Large Hadron Collider and the Human Genome Project. DPI tends to flourish when public infrastructure creates space for private innovation. Open, modular technology allows startups and enterprises to build and innovate new products.

3. Identify use cases through cost–benefit analysis

Early use cases for DPI should be chosen on the basis of feasibility and national priorities. A coordinated analysis on the benefits of DPI should weigh long-term efficiency gains against upfront infrastructure costs. This will require coordination among significant government budget holders, digital regulators and sector-specific agencies to ensure accountability and alignment of incentives. For countries currently lagging behind DPI leaders in digital public service provision, there are significant opportunities in simply replicating existing DPI successes, where similar contexts exist. However, DPI is diverse, and examples like the Ukrainian government’s drive for services capable of functioning during conflict show that different governments will have differing priorities for building new digital foundations.

4. Plug into global DPI networks

A major strength of the DPI approach is that it enables countries to foster interoperability and collaborate across borders while preserving national autonomy. These alternative diplomatic channels are vital levers in the current, shifting global governance landscape. To boost interoperability and strengthen partnerships, once DPI capacity is established, countries should participate in emerging regional or sectoral DPI communities of practice that help set international technology standards that reflect national strategic priorities or values. Such initiatives include the Global Digital Compact, 50-in-5 initiative, Global Digital Health Certification Network, and Digital Public Goods Alliance.

Recommendations in practice

The UK government’s 2025 announcement of mandatory digital IDs demonstrates the political commitment needed for DPI implementation.⁷⁰ This initiative could leverage existing institutional capacity, notably ARIA and the GDS, both of which have already built expertise in digital infrastructure. This capacity would need additional investment and a procurement process guided by the Digital, Data and Technology Playbook (UK government guidance for digital projects) to ensure the adoption of a suitable, safe public–private partnership for digital ID.⁷¹

The government has identified a specific use case: right to work checks. While this provides a concrete starting point, for some, this limits the potential of the technology, which could have also extended to delivering better public services

⁷⁰ Prime Minister’s Office (2025), ‘New digital ID scheme to be rolled out across UK’, 26 September 2025, <https://www.gov.uk/government/news/new-digital-id-scheme-to-be-rolled-out-across-uk>.

⁷¹ Digital, Data and Technology Profession (2022), ‘The Digital, Data and Technology Playbook’, 28 March 2022, <https://www.gov.uk/government/publications/the-digital-data-and-technology-playbook>.

in other areas such as healthcare or taxation. Ensuring interoperability across departments and embedding data sharing could help future-proof the digital ID for further expansion, and potentially improve delivery of public services, increase efficiency, reduce duplication and enhance governance. With a more digitally connected government, the UK could better understand and address complex societal challenges, such as social care, health disparities, quality of education and homelessness.

Public concerns around data privacy and government oversight must be addressed. This could be done through a system similar to the Data Empowerment and Protection Architecture, an India-developed proposal for a joint public-private oversight instrument that enables individuals to share their personal data with service providers and third parties through personalized and specific consent controls.⁷² In India, the country's instant payment sector is a diverse and competitive market, with platforms like Google Pay, PhonePe, Amazon Pay and BHIM, all offering instant, real-time transfers. These providers all have different data and privacy policies dictating the use of user's data. The government has recognized this shortcoming. By consolidating and pooling consent mechanisms, governments, like those in India and the UK, could introduce a system with familiar and standard wording that is more likely to be trusted by users. By carefully designing systems with users' needs, consent and interactions in mind, governments can create stronger foundations for trusted private sector partnerships and enable more advanced digital infrastructure, like secure, privacy-protected e-commerce platforms. This advancement would enable providers to reduce fraud and streamline processes while giving consumers greater control over their personal information.

Rather than developing digital ID systems in isolation, the UK should actively engage with the global DPI community. As this paper emphasizes, countries can learn lessons from existing DPI implementation, adopting proven vendors and building blocks, and participating in international standards-setting frameworks. The paper provides numerous examples of successful DPI deployments and collaborative initiatives that UK policymakers should study and engage with. This approach would help the UK avoid costly mistakes, accelerate implementation and ensure that a UK digital ID system benefits from best practice and adheres to international standards.

⁷² NITI Aayog (2020), *Data Empowerment And Protection Architecture*, August 2020, <https://www.niti.gov.in/sites/default/files/2023-03/Data-Empowerment-and-Protection-Architecture-A-Secure-Consent-Based.pdf>.

Annex 1

Existing digital solutions

This list is not exhaustive, but is an illustration of the diverse selection of existing digital building blocks and public goods in use worldwide in digital public infrastructure approaches.

Country/organization	DPI name	Online resource
India	Aadhaar	https://uidai.gov.in
Worldwide (but supported by France)	CurrentAI	currentai.org
Denmark	Digital Post	https://en.digst.dk/systems/digital-post
Ukraine	Diia	https://expo.diia.gov.ua
EU	eIDAS	https://digital-strategy.ec.europa.eu/en/policies/discover-eidas
EU	EU Digital COVID Certificate	https://commission.europa.eu/strategy-and-policy/coronavirus-response/safe-covid-19-vaccines-europeans/eu-digital-covid-certificate_en
EU	EuroStack	https://eurostack.eu
Sri Lanka	GOVPAY	https://govpay.lk/en
UK	gov.uk	https://www.gov.uk
Saudi Arabia	The Hajj health card	https://www.who.int/news/item/21-10-2024-who-global-network-expands-digital-health-certification-for-hajj-pilgrims
Brazil	Identificação Civil Nacional	https://www.gov.br/governodigital/pt-br/legislacao/legislacao-identificacao-nacional-civil
Kenya (and now elsewhere)	M-PESA	https://www.m-pesa.africa
Bangladesh	myGov	https://www.mygov.bd
Togo	NOVISSI	https://documents1.worldbank.org/curated/en/099751009222330502/pdf/IDU0e83f857301ff1047bf082710a8d21ddf42c3.pdf
Brazil	Pix	https://www.bcb.gov.br/en/financialstability/pix_en
Peru	State Interoperability Platform (PIDE)	https://www.gob.pe/741-plataforma-de-interoperabilidad-del-estado-pide
Singapore	Singpass	https://www.singpass.gov.sg/main
Sri Lanka	Sri Lanka Unique Digital Identity (SLUDI)	https://www.icta.lk/projects/digital-government/sludi
Estonia	X-Road	https://x-road.global

About the authors

Rowan Wilkinson is a research associate in the Digital Society Programme at Chatham House, where she conducts research on digital public infrastructure, the information space, technology sovereignty and AI governance. Her expertise lies at the intersection of technology, humanitarianism and international development.

Alex Krasodomski leads the Digital Society Programme at Chatham House. His research focuses on AI, emerging technology and centres of technological power. He leads work aimed at strengthening state capacity and cooperation, identifying feasible paths towards global technology governance, and on routes to market for public technology.

Isabella Wilkinson is a research fellow in the Digital Society Programme at Chatham House. Isabella's work generally covers security and governance issues relating to cyberspace and technology, including international cyber and technology governance and processes, the online information environment, and advancing responsibility and inclusivity.

Acknowledgments

We are grateful to Co-Develop and, in particular, Lindsey Crumbaugh and Sourav Das, for their support for this project.

We would like to express our sincerest gratitude to the anonymous peer reviewers for their insights and constructive feedback. This paper also draws on roundtables and interviews held under the Chatham House Rule in 2025. The authors are grateful to all those who participated, giving their time and valuable comments.

Thanks are due to Chatham House's Publications team and the staff of the Global Governance and Security Centre for their support, particularly Bianka Venkataramani, Jake Statham and Michael Tsang.

All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical including photocopying, recording or any information storage or retrieval system, without the prior written permission of the copyright holder. Please direct all enquiries to the publishers.

Chatham House does not express opinions of its own. The opinions expressed in this publication are the responsibility of the author(s).

Copyright © The Royal Institute of International Affairs, 2025

Cover image: A vegetable vendor waits for customers and displays a barcode for Paytm, an Indian mobile digital payment platform, at a market in New Delhi.

Photo credit: Copyright © Sajjad Hussain/AFP/Getty Images

ISBN 978 1 78413 660 4

DOI 10.55317/9781784136604

Cite this paper: Wilkinson, R., Krasodomski, A. and Wilkinson, I. (2025), *The case for expanding digital public infrastructure: How open, scalable technology can serve citizens, preserve sovereignty and save money*, Research Paper, London: Royal Institute of International Affairs, <https://doi.org/10.55317/9781784136604>.

This publication is printed on FSC-certified paper.
designbysoapbox.com





Independent thinking since 1920



**The Royal Institute of International Affairs
Chatham House**

10 St James's Square, London SW1Y 4LE

T +44 (0)20 7957 5700

contact@chathamhouse.org | chathamhouse.org

Charity Registration Number: 208223